Forecast the price of chemical products with multivariate data

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***Abstract*—Sales price of staple commodities plays an important role in human life and reflects production and sales of enterprises, so predicting the price accurately is of great significance. The price of chemical products has the characteristics of time series, nonlinear, unstable, etc, and has relationship with multiple variables which are affected by seasons, national policy and macro-economy. Therefore, predicting their price has become a challenging task. In this paper we propose a new prediction algorithm that exploits multivariate data with analysis including crawled web data related to chemical products and expert experience data. History data is first disposed and analyzed to build statistic and machine learning forecasting models. Then sentiment analysis is performed based on related data crawled from the internet measured by text analyzing techniques. Finally expert experience on forecasting the price is used to optimize the prediction results. We use methanol as an example to evaluate the accuracy of prediction results tracked for eight months, the MAPE (average absolute percent error) of our method is 2.91% better than other models. Compared with traditional prediction models, our model based on multivariate data has higher accuracy.**

Keywords-chemical products; time series; multivariate data; sentiment analysis; expert experience

I. INTRODUCTION

Predicting the price of staple commodities such as oil, agricultural commodities, chemical products etc, has become more and more important due to the growth of economy and the competition among enterprises. Accurate prediction provides effective measurement for company management and reflects economy situation indirectly.

Although there are quantitative and qualitative prediction methods, it is very difficult to predict the price of chemical products. The quantitative methods include time series prediction and causality analysis while the qualitative prediction methods mainly focus on human experience and subjective judgment, but they seem to be very difficult to predict the price with rapid fluctuation. In recent years, machine learning has been widely used to predict price for its ability in intelligence learning, including back propagation (BP) neural network [1], wavelet neural network (WNN) [2] and support vector machine (SVM) models [3] etc. Meanwhile machine learning models need variables as input because different variables may lead to quite different results. However, there are many variables affecting the price of chemical products, such as product supply and demand, macro-economy, national policy, financial crisis and so on. At the same time, the price of chemical products was non-linear, unstable and fluctuant, which makes predicting the price of chemical products a more difficult task. The challenges of prediction focus on:

- How to consider those influence variables. The variables from a variety of raw data sets are redundant and very difficult to choose. Different input variables will lead to different effects on the prediction results [4].
- How to deal with the fluctuate price that affected by macro-economy, psychologic factor, emergency etc. The quantifiable factors cannot reflect the price's fluctuation intuitively.
- How to integrate crawled web data related to chemical production price and expert experience into prediction model meanwhile optimize input parameters and prediction results.

Motivated by the above, this paper proposes a prediction method based on multivariate data, including the history statistical data, web comments data, and experience of experts. What's more, feedback adjustment is introduced to optimize the model. We tracked the prediction results for eight months and succeeded in predicting the price. We evaluated our prediction model compared with three other models. Experimental results demonstrated that our method achieved statistically significant improvements in precision. The contributions of the paper are as follows:

- The dimension reduction method is proposed, instead of treating the variables equally. After clustering the variables into different categories we use stepwise regression method [5] to find the representative variables.
- Different statistical and machine learning models are used to predict methanol price based on selected variables. Meanwhile we find out optimal models for

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predicting the price of chemical products depending on their properties.

- A multivariate data-based approach is proposed, which combine emotion analysis based on the data crawled from the Internet and experts' experience from questionnaires by investigation with the optimal models for predicting.

The rest of the paper is organized as follows. Section 2 presents the related work. Section 3 discusses forecasting by analyzing history data. Optimization based on analysis and expert experience in Section 4. Section 5 explains informal evaluation. Finally is the summary of the paper.

II. Related Work

There have been some works about forecasting the price of different markets' products, they used different prediction models, and the price of those products is affected by many variables and has the characteristic of unstable and fluctuation. Lasses et al [6] used a common factor approach to forecast the US housing price. Eric and Alberto [7] explained the evidence of predictability in US residential and commercial real estate markets. The multivariate linear regression analysis technique and the time series analysis technique are used by Ming [8] to forecast the coal price. It turned out that the improved traditional partial least-squares regression could forecast the coal price [9]. Meanwhile, the autoregressive models that incorporate information from numerous economic variables can forecast house price accurately [10]. Shen [11] used the machine learning algorithm that exploited the correlation among global stock markets and various financial products to predict the stock market.

While there exited different methods that were suitable for short term price forecasting (STPF), and they could reduce the fluctuation of the price and smoothed out volatility. The original ARCH model has been commonly used to forecast changes in the volatility of financial time series [12] in 1994. Morana [13] proved that the GARCH is suitable for the short-term forecast of the oil in 2001. While Khosravi [2] proposed neural network and Generalized Auto Regressive Conditional Heteroscedasticity (GARCH) models to forecast electricity price and estimate their variance in 2013. In the same year Gabralla [14] applied the machine-learning approaches and L Yu [15] used an EMD-based neural network to forecast crude oil prices.

An adaptive WNN for short-term price forecasting in the electricity markets was proposed by Pindoriya [16] in 2008 and the BP Neural Network [1] was applied for the short term load forecasting in 2009. They both have the ability of nonlinear mapping, self-learning and adaptive, and easy to remember the relationship between inputs and outputs, which are suit for forecasting the fluctuate price.

Recently, with the increase of big data, some works have used different data source for prediction. Wu [17] integrated multiple data that based on news articles, historical stock price and market state to predict stock. Puniyani [18] proposed NP-MuScL to estimate a gene interaction multiple data sources from network to predict the relationships between the genes. While Zheng [19] used the real-time data from the monitor station and fine-grained air quality information to forecast the air quality, the data plays an important role in forecasting and improves the accuracy of prediction. From the above works, we believe that data analysis will exert positive effects on prediction.

III. Forecasting by Analyzing History Data

As we all know, noisy and unreliable data will affect the accuracy of forecasting, and redundant and irrelevant data may lead to inaccurate predicted results [20]. Meanwhile Guyon [21] has mentioned that when dealing with many features, some of them are interdependent and convey the same information. Therefore, preprocessing the data of chemical products is necessary for prediction.

A. Preprocessing of History Data

In this section, data cleaning, normalization, classification and selection are needed for the provided raw data. There are too many variables related to those chemical products, but only a few of them are valuable. In this paper, we have done something especially for our classification and selection. We use methanol as an example. The data is provided by one Chemical Industry Institute, which professionally engages in the chemical industry competitive research, monitoring and analysis. The data includes a total of 84 records from Jan.2008 to Dec.2014, and each record has 21 properties: Cost and Freight (CFR) China, Oil price, Coal price, PMI, etc.

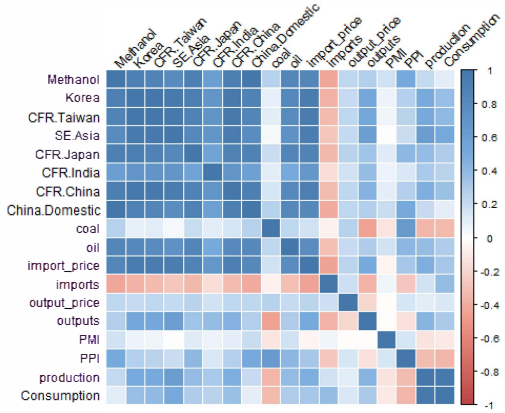

Figure 1. The correlation between the methanol and other variables represented by different colors. From 1 (blue) that represents the most positive correlation to -1 (red) the most negative variables.

The variables recorded by the company have correlation with each other (As shown in Fig.1). Apparently, some variables have high relationship with the methanol price whose value is as high as 0.9, and other variables have relatively low relationship with it. We will select the significant variables by the following two steps:

1) Data cleaning and normalization. There are statistical errors mostly from data acquisition and artificial statistics needed to be handled. On one hand we use data smoothing to smooth the incorrect data in order to improve the stability and accuracy of the model; on the other hand,

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we reserve and record the abnormal data affected by unexpected incidents, those data will influence the stability and accuracy of the prediction model.

The data may have different values, some of which range from 1 to 100, but some others range from 1000 to 5000. Therefore, normalization is necessary. There are two normalization methods: min-max normalization and z-score normalization. We use the second method here:

$$Z(x) = \frac{x - mean(x)}{StaDiv(x)} \tag{1}$$

2) Classification and variables selection. The price of chemical products is affected by many unstable variables. In order to avoid degrading the performance of the prediction, we will remove some unnecessary variables from the input variables to reduce the dimensions of factor space [4].

Here the quantitative variables provided by the company are firstly considered (As listed in Fig.1), and the non-quantifiable factors will be introduced in Chapter 4. Based on the properties of the chemistry products, we divide the variables into different categories after a long time investigation, analysis and discussion. The variables of methanol price are divided into six categories (As depicted in Fig.2), which cause the fluctuation of methanol price.

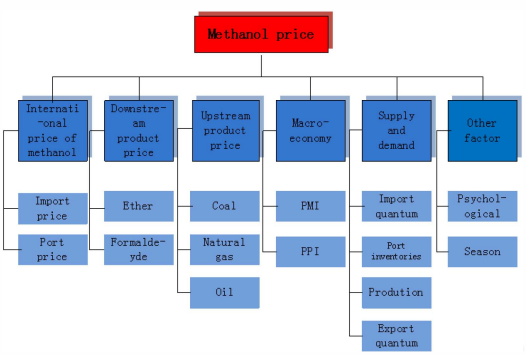

Figure 2. A typical chart to present the complex relationship of the variables of the methanol price. The variables of methanol can be divided as best as possible into six categories

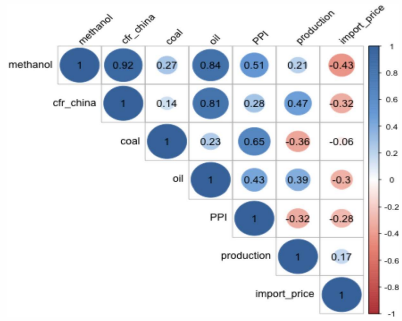

Figure 3. Schematic description of the correlation of methanol price with other variables, the right color bar reflects the size of correlation, and the final correlation degree with other variables.

Through the above analysis, we carefully examine the impacts those variables have on the methanol price. We use stepwise regression [5] to obtain useful variables, which will be verified by two regulations:

- The variable that has the low correlation with the methanol will be removed from the final results.
- The variable must have integrity and availability.

Though those disposals hereinbefore, the final variables can be obtained. It is turned out that the most relevant variable is CFR china. Fig.3 represents the final variables: port price, coal price, oil price, PPI, production and import are the relatively important variables, while the import price has negative effects on it.

B. *Prediction based on Quantitative Factors*

It is difficult for us to predict the price of chemical products due to its fluctuation, complexity, and irregularity. The fluctuation shows that the existence of a wide range of price, which is related to the changing of macro-economy. The complexity is that it depends on many unquantifiable factors, such as national economic and environment.

Faced with those features of the price, we use GARCH model [13] that can deal with the persistence volatility price and the machine learning models which can learn the feature of the history data, such as BP neural network and WNN models. For the self-correlation feature of methanol, we use the Auto-Regressive and Moving Average Model (ARMA) [22], exponential smoothing (ES) [23] and seasonal exponential smoothing (SES) [24] models. We will predict the price and make comparison between the linear and nonlinear models.

Then we examine the performance of different models and apply the various evaluation criteria to evaluate prediction results. The GARCH model is better when dealing with fluctuate, complex and irregular price of methanol than others (Fig.4, Fig.5), which has the best fitting degree and the smallest MAPE, and apparently the deviation rate and the MSE are better than linear models. Though the BP and WNN models are fit for the STPF, their uncertain initialization parameters lead to the unstable prediction results. Up to this point, we only consider the quantifiable factors, and there are other unquantifiable factors from crawled methanol price related web data and expert experience that we hope to improve prediction accuracy.

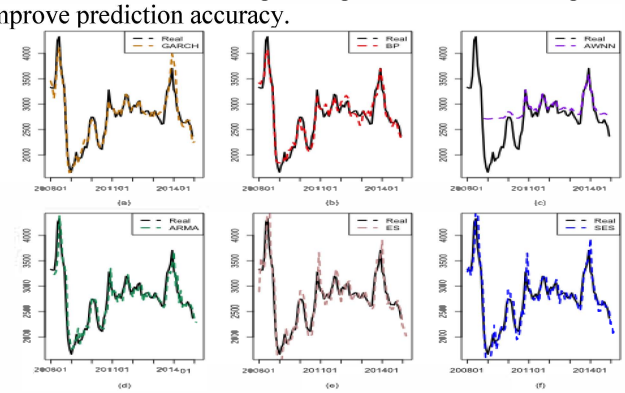

Figure 4. The different of curve fitting : (a) GARCH model (b) BP neural network model (c) WNN model (d) ARMA model (e) ES model (f) SES model.

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Figure 5. The evaluation criteria of the fitting, including MAPE, Goodness of fit (deviation degree of real value and predicted value), Deviation rate, and MSE (mean squared error).Optimization based on Sentiment Analysis and Expert Experience.

IV. Optimization Based on Sentiment Analysis and Expert Experience

In the above work, we select important factors from history data and predict methanol price. But the price of chemical products always fluctuates according to the season, human emotion and economy whose effects seem very difficult to quantize. For the next month, there may be some unstable factors such as the changing of futures' prices and politics will affect expected price. In this section, we attempt to optimize our prediction model by using a novel algorithm which can calculate emotion value from the crawled web, and find functions to calculate the expert experience.

A. Sentiment Analysis on Web Data

1) Data Crawling

In this paper, the Web Crawler [25] is used to fetch the pages about related information of chemical products. For example, we use three URL sites that publish news about methanol price every day to crawl a large amount of commiting documents from Oct.2010 to Dec.2014, an average of 44 documents per month, more than 2200 in total.

This chapter explains the sentiment analysis in detail. Firstly, data preprocessing [26] should be done to reduce data redundancy. Then we will calculate the sentiment value that will reflect the positive or negative of the document accurately. Finally we will compare the emotion value with the fluctuant price of chemical products (As shown in Fig.6).

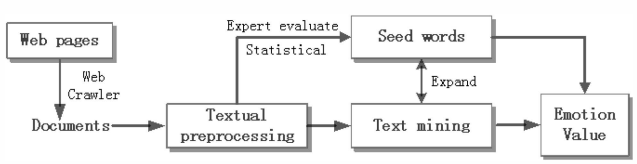

Figure 6. Overview of the sentiment analysis of web data.

2) Opinion Mining

After text preprocessing including cleaning, filtering, classification, selection and word segmentation, each document can be presented as a list of words. Some of the words have sentiment information for methanol price. What we want to do is to give each document an sentiment value based on its words in order to predict the methanol price next month. So identifying the words with sentiment and evaluating their levels is a significant but challenging task. For sentiment analysis, there are some Chinese emotion word dictionaries such as SentiWordNet [27] and NTUSD [28], which mainly focus on people's feeling and emotions. However, there are few emotion words dictionaries for chemical products, so we need to construct a suitable dictionary for chemistry products.

Firstly, we need to get the Seed Words. After splitting sentences from all the crawled documents, we seek out the high-frequency words, where we invite professional experts to pick up some words that with strong emotion (positive and negative), then confirm their values. The value of the chemistry products words mainly based on Mainte Taboada [29] who set the scores ranging from -0.5 to +0.5, ant it represents the negative to positive emotion, having eleven categories. Then, we use Continuous Bag-of-Words Model [30] [31] which is the improvement of Neural Network Language Model (NNLM) [32] to enlarge seed words, the model use the Softmax [33] algorithm and combine with Huffman Coding. The probability of one word is based on the context, the function is as follows:

$$P(w_i \mid context) = P(w_i \mid w_{i-k}, w_{i-k+1} \cdots, w_{i-1}, w_{i+1}, \cdots, w_{i+k}) \quad (2)$$

Where P represents the probability of word w_i in the context. We use machine learning method rather than manually tagging to build our polarity classifiers and to expand our emotion dictionary, meanwhile we deal with some errors artificially. The seed enlarging algorithm is described as Fig.7.

Algorithm: *FindSeedWords(SW)*
Input: *Words* (most positive and negative)
Begin
  For each *word* in *Words*
    Use Continuous Bags-of-Words model
    to obtain *wordlist*;
    for each w_i in *wordlist*
      if ($w_i.correlation >= 0.9$)
        $w_i.score = w_i.correlation * word.value$;
        add w_i to *Words*;
  Calculate the value of each Document.
End
Output:*Value_Doc*

Figure 7. Seed enlarging algorithm

Based on the above work, we build our chemical products dictionary so as to calculate the emotion value of each document, where the emotion value for each month is got. Md_i represents the number of days in month i , the function is as follows:

$$Month_value_i = \frac{\sum_{j=1}^{Md_i} v_i}{Md_i} \quad (3)$$

Fig.8 describes the trend of emotion data and the fluctuation of methanol price. It clearly concludes that the

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emotion value has the same trend with the methanol's fluctuation. Naturally, analyzing web data has great significance in predicting the price of methanol. We can summarize the web data to give a subjective tendency for methanol price. Though the emotion has a point conforming to the price trend, the delay attribute is also remained.

Figure 8. This diagram reflects the trend of emotion data and the fluctuation of methanol price, in which the positive and negative emotion can be reflected. It reflects the emotion data from web sites and increment of methanol from Oct, 2010 to Dec, 2014.

B. *Experience on Questionnaires evaluation*

Anyway, the experience of experts is also valuable. Here we use the expert questionnaires aiming at forecasting for data mining, from which capture and analyze important information to improve accuracy and reliability of the prediction models. We can perform it through the next steps:

- Acquiring the latest methanol questionnaires judged by intelligence agents;
- Evaluating and selecting those questionnaires to identify experts' judgment;
- Soliciting ideas for future trend.

1) *Procedure and Participants*

During the period of forecasting methanol price, three different questionnaires are designed according to three kinds of companies, including methanol's traders, vendors and downstream companies. Each of questionnaires has more than 20 questions and each company has only one questionnaire.

To a certain degree, the experts should have marketplace analytical ability and know about the national politics and macro-economy. After investigation, we find that the traders have diverse sales volume, the vendors have different accommodation, and the downstream enterprises have different demands for methanol. In the beginning, we think they have the same impact on the methanol. After investigation, we find that the traders' and vendors' experience value are more important than downstream enterprises', so we determine their percentage by their volume information and experimental result, the percentage of questionnaires: Trader is 40%, Supplier is 40%, and Downstream is 20%.

2) *Questionnaire assessment*

Instead of obtaining the questions' answers hastily, we give weight to the questionnaire according to seriousness and reliability of the judgment. During the investigation, from Apr, 2014 to Dec, 2014, we find that the experience of experts is based on two main characteristics: working year, understanding of the industry. The longer the expert stays in the chemical industry, the more reliable experience. The better an expert understand the market, the more experience we think he has. Those characteristics have a potential impact on the methanol. We care more about the quality of questionnaires than the quantity.

TABLE I. WEIGHT OF EACH QUESTIONNAIRE

	Level	**Weight**
Expert experience	Rich(%)	60%
	Common(%)	30%
	Poor(%)	10%
Expert's working years	Years ≥ 15(%)	40%
	15>Years ≥ 7(%)	30%
	Years<7(%)	20%

Finally we summarize the experts' experience and working years, and give different weights of those questionnaires (Table 1). The questionnaires' value was quantified based on the regulation in above.

V. INFORMAL EVALUATION

Having tracked the data from Apr.2014 to Dec.2014, we evaluated the prediction models (As illustrated in Table.2). The price forecasted at the beginning of each month is compared with the real price counted at the end of the month. The error of GARCH model is as low as 0.0304, and the error of ARMA combined with GARCH for Two-step and Three-step are 0.0465 and 0.0595. However, STPF only forecast the price of next month while LTPF can give a trend in the coming two months. The prediction results will affect the decision of a company. Improving accuracy of the prediction may obtain enormous profits.

TABLE II. WEIGHT OF EACH QUESTIONNAIRE

Evaluation criteria	**one month prediction**	**two month prediction**	**Three month prediction**
GARCH	0.0304	-	-
BP	0. 0360	-	-
WNN	0.0401	-	-
ARMA	0.0449	-	-
ES	0.0385	0.0680	0.1100
SES	0.0420	0.0636	0.0937
GARCH & ARMA	-	0.0465	0.0595
GARCH & ES	-	0.0477	0.0661
BP & ARMA	-	0.0527	0.0672
BP & ES	-	0.0596	0.0707
WNN & ARMA	-	0.0591	0.0659
BP & ES	-	0.0624	0.0826

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A. *Our Prediction Model*

After using different statistical models and machine learning methods, we can find out the best model for the methanol is GARCH combine with ARMA model. The errors of the statistical models are mentioned in Table 2. Our model is based on the GARCH (we also do an experiment with other models), then combine with the emotion value from the web data and expert experience in order to reduce the forecasting errors of fluctuant price and improve the accuracy of forecasting (Fig.9), which can give the STPF of chemical products' price. We also use ARMA to give the LTPF.

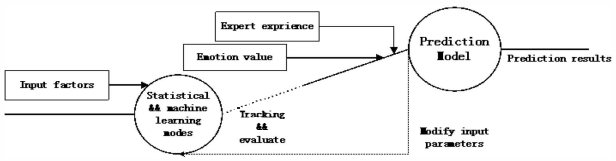

Figure 9. The statistical method combining web data and expert experience.

The function of modify prediction results is as follows:

$$FR(i) = \alpha \bullet EV(i) + \beta \bullet PV(i) + MV(i) \tag{4}$$

Where $FR(i)$ is our prediction result of month i. $EV(i)$ and $PV(i)$ stand for the value of emotion, experience of month i, and $MV(i)$ is the models' prediction value. α, β are the parameters. We use the $EV(i)$ and Md_i to modify the model's input parameters, α and β.

B. *Prediction Evaluation*

With the increase of data, we use the least squares to fit a generalized linear model, it will return the best estimate parameters (α, β), which are available to forecast the value of month i. And we compare the real price with our prediction results every month to optimize the model and adjust the parameters.

Fig.10 shows that the average MAPE of our method that tracked from Apr, 2014 to Dec, 2014 is 2.91% perform better than other models (As shown in Table.2), it seems more adapt for volatile price than traditional models. Although the BP and WNN models seem to capture the real tendencies intuitively, we can find out that our method predict value more close to the real price.

As we all know, the falling oil price affected by international variables mainly leads to the methanol' price falling in 2014, in the same time the poor performance of chemical products' market leads to negative emotions. From Fig.10, our prediction that considers the macro-economy and national policy performs better in the fluctuating price than the pure machine learning models. Combined with sentiment analysis and experts value, our model' prediction result is fit for the real price and the accuracy and reliability are improved. Each month, we will send the results back to the company and give them suggestion of next month.

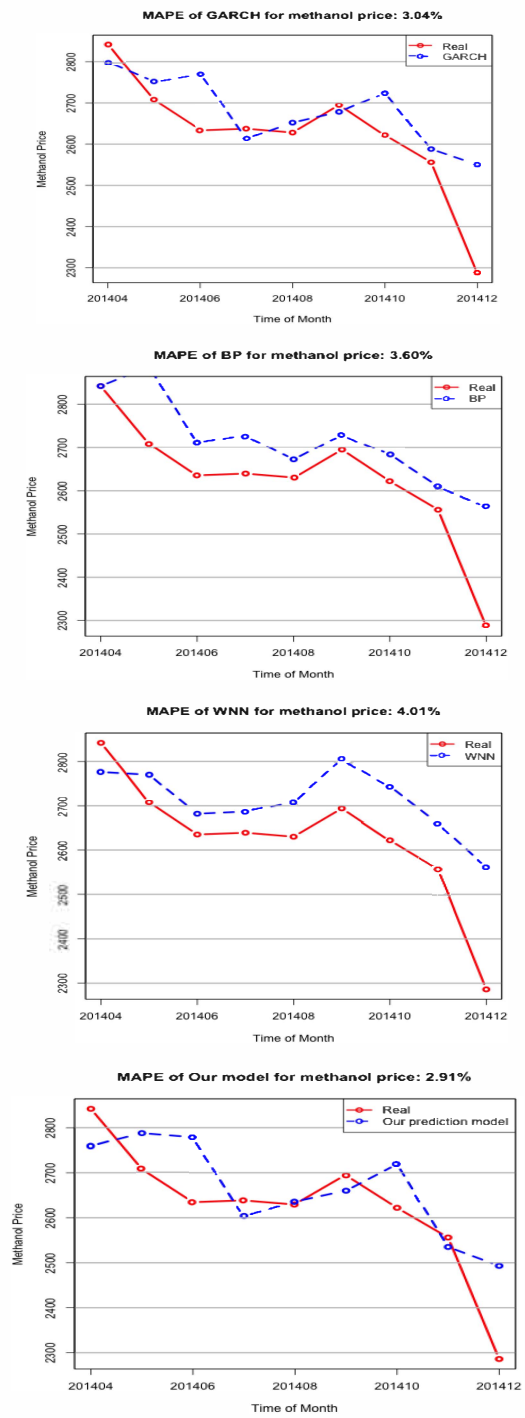

Figure 10. The compare of real data with the prediction results of different models, it reflects that our model with emotion and experience is better than others.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, a novel prediction model for the price of chemical products combined with multivariate information is proposed. We have summarized the correlation variables for the chemistry products while considering their feasibility.

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Our methods can predict the methanol price with web data and expert experience from the typical companies. What's more, it's a general method that is fit for predicting the price of other staple commodities as well. From the analysis of prediction results we can find that our method mixed with web data and expert experience has more accurate results. But there also exits continuous and sharp fluctuation that we cannot predict accurately.

In the future, with the increasing of web data, we will search more useful information to improve the accuracy of the prediction results. For sentiment analysis, we will also record the commentator information and regularize their weights dynamically next to give different weight for the news document. In all, the future predicting will become more and more accurate under the influence of the increasing web data and abundant expert experience.

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